



Diesel Generator Set

mtu 12V4000 DS2000

380V – 11 kV/50 Hz/standby power/fuel consumption optimized
12V4000G84F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 1970 kVA - 2080 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- Fuel consumption optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)		
Manufacturer	mtu		Total oil system capacity: l	260	
Model	12V4000G84F		Engine jacket water capacity: l	160	
Type	4-cycle		Intercooler coolant capacity: l	40	
Arrangement	12V		Combustion air requirements		
Displacement: l	57.2				
Bore: mm	170			Combustion air volume: m³/s	2.0
Stroke: mm	210			Max. air intake restriction: mbar	50
Compression ratio	16.4		Cooling/radiator system		
Rated speed: rpm	1500				
Engine governor	ECU 9			Coolant flow rate (HT circuit): m3/hr	56
Max power: kWm	1750			Coolant flow rate (LT circuit): m3/hr	30
Air cleaner	dry		Heat rejection to coolant: kW	630	
Fuel system			Heat radiated to charge air cooling: kW	340	
			Heat radiated to ambient: kW	75	
			Fan power for electr. radiator (40°C): kW	38	
Maximum fuel lift: m	5		Exhaust system		
Total fuel flow: l/min	16				
Fuel consumption ²⁾	l/hr g/kwh			Exhaust gas temp. (after turbocharger): °C	470
				Exhaust gas volume: m³/s	5.1
			Maximum allowable back pressure: mbar	85	
			Minimum allowable back pressure: mbar	30	
At 100% of power rating:	413.3	196			
At 75% of power rating:	300.5	190			
At 50% of power rating:	208.7	198			

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	fuel consumption optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S6 (Low voltage Leroy Somer standard)	380 V	1664	2080	3160	1624	2030	3084
	400 V	1664	2080	3002	1624	2030	2930
	415 V	1664	2080	2894	1624	2030	2824
Leroy Somer LSA52.3 S7 (Low voltage Leroy Somer oversized)	380 V	1664	2080	3160	1624	2030	3084
	400 V	1664	2080	3002	1624	2030	2930
	415 V	1664	2080	2894	1624	2030	2824
Marathon 744RSL7091 (Low voltage Marathon)	380 V	1576	1970	2993	1576	1970	2993
	400 V	1624	2030	2930	1608	2010	2901
	415 V	1608	2010	2796	1608	2010	2796
Marathon 744RSL7092 (Low voltage Marathon oversized)	380 V	1576	1970	2993	1576	1970	2993
	400 V	1624	2030	2930	1608	2010	2901
	415 V	1608	2010	2796	1608	2010	2796
Marathon 744RSL7092 (Low voltage Marathon engine output optimized)	380 V	1648	2060	3130	1616	2020	3069
	400 V	1656	2070	2988	1616	2020	2916
	415 V	1640	2050	2852	1616	2020	2810
Marathon 1020FDH7096 (Medium volt. marathon)	11 kV	1656	2070	109	1608	2010	105
Leroy Somer LSA53.2 VL7 (Medium volt. Leroy Somer)	11 kV	1664	2080	109	1624	2030	107

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.